

Reboot the Earth

Global Tech Challenge 2026



Doha, Qatar



September 23-26

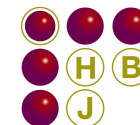


UNITED NATIONS
OFFICE OF INFORMATION AND
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Food and Agriculture
Organization of the
United Nations

Carnegie
Mellon
University
Qatar



مركز حمد بن جاسم
لتعليم علوم الحوسبة
HAMAD BIN JASSIM CENTER
FOR K-12 COMPUTER SCIENCE EDUCATION

Meet your Hosts



Mithusa Kajendran

Open Source Adoption Consultant,
United Nations Office of
Information and Communications
Technology



Eduardo Feo-Flushing

Assistant Teaching Professor,
Computer Science
Carnegie Mellon University in
Qatar



Nour Elhouda Tabet

Outreach Manager,
Computer Science
Carnegie Mellon University in
Qatar

You are
already a
part of
something
bigger!



RTE NYC



RTE CMU-Rwanda



RTE India



RTE Doha 2025



A Word from our Partners

Omar Mohsine

UN Open Source Coordinator



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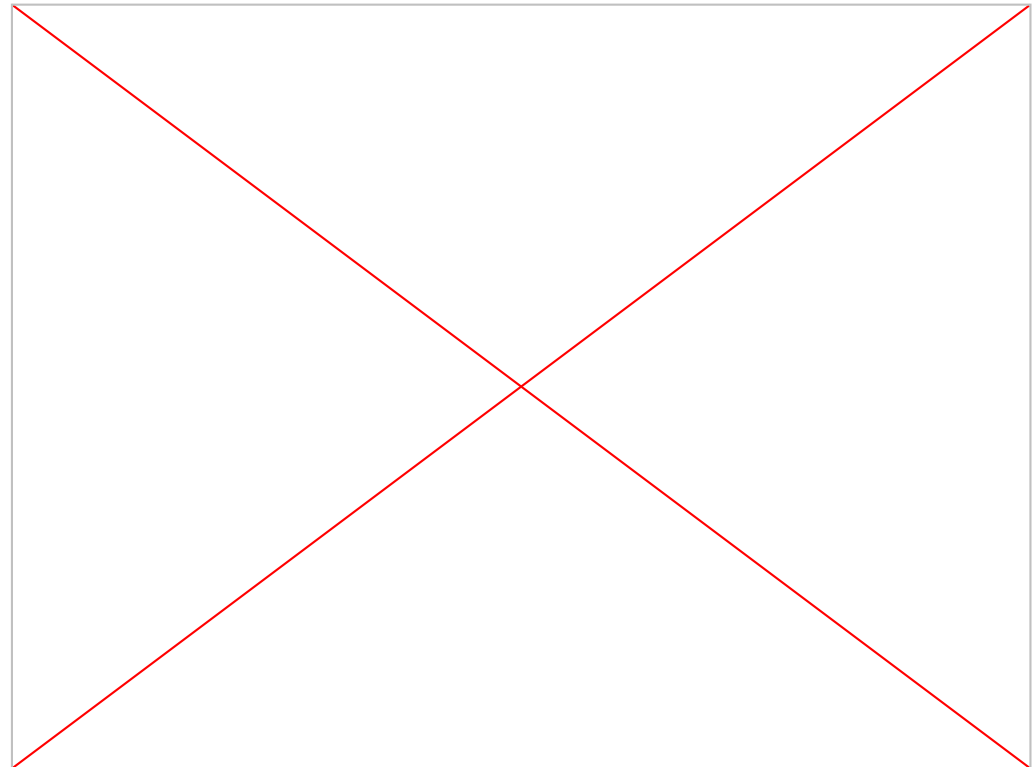
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UN Open Source Coordinator



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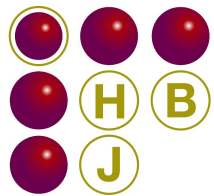


A Word from our Partners

Nour Tabet

Outreach Manager

Hamad Bin Jassim Center at CMUQ



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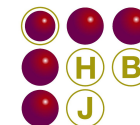


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Getting Help

Join our Slack workspace

If you have not joined yet, check your email for the invitation. If you experience any issues, speak with one of our volunteers.

Connect with our mentors

Watch Slack for mentoring session announcements. Sessions will take place in the walkway and are available on a first-come, first-served basis. You can sign up in advance through Slack.

Attend a workshop

We have workshops on sustainability, open source, technical presentations, and advanced AI topics. Sessions run from tomorrow afternoon through Saturday morning. Attendance is optional but highly encouraged. Check Slack for the schedule and updates.

Participant Handbook



Meet your Mentors



Vipul Siddharth

Open Source Innovation at
UNICEF (Online Mentor)



Eduardo Feo-Flushing

Assistant Teaching Professor
CMU-Q



Gianni Di Caro

Teaching Professor
CMU-Q



S. Duygu Sever

Visiting Assistant Professor
CMU-Q



Kira Dreher

Associate Dean, Community Excellence
/ Associate Teaching Professor
CMU-Q



Abrar Abir

Research Specialist
Sidra Medicine

Deep Chandra
Software Engineer, UBS

Moamin Ibrahim
Research Assistant, QCRI

Nada Sulaiman
Software Engineer, Snoonu

Abdelrahman Abdelnaby
PhD Candidate, Saarland University

Sunaya Upadhyay
Research Assistant, QCRI

**Stay tuned on Slack for more
information about the mentorship
sessions and workshops.**

The Rules

Open Source, Open Data, Open Everything

We don't just love Open Source, we require it, for our future's sake. We also encourage you to leverage digital public goods as much as possible

Original work and IP

We only want your work, and any open source dependencies you use with proper attribution

Always be respectful and inclusive

Abide by the Code of Conduct at all times, and foster a collaborative environment

Inter-team Collaboration

Though we love collaboration across teams, each team must submit its own solution

Code Quality

Follow commonly accepted coding standards and don't forget to document your code

The Challenges

**Challenge 1:
AI for Site/Facility
Suitability and
Crop/System
Recommendation**

**Challenge 2:
AI for Cold-Chain
Monitoring and
Food-Loss Reduction**

Choose one challenge!

Challenge 1

AI for Site/Facility Suitability and Crop/System Recommendation

Example features



Site Suitability Intelligence

Use GIS data on solar exposure, wind and dust risk, soil salinity, water availability, and market access to assess the suitability of potential farming locations.



Crop and Production System Recommendation

Match each site with suitable crops and production systems, including open field, greenhouse, hydroponic, and vertical farming, based on environmental conditions, resource needs, market demand, and budget.



Farm Investment Scenario Planning

Compare expected yield, resource consumption, CapEx, OpEx, revenue, and payback scenarios to evaluate the financial feasibility, requirements, and risks of different production options.

Challenge 2

AI for Cold-Chain Monitoring and Food-Loss Reduction

Example features



Cold-Chain Data Integration

Combine temperature and humidity readings with product information, location, handling history, and transportation and storage records.



Spoilage and Shelf-Life Prediction

Use AI to identify cold-chain anomalies, estimate spoilage risk, and predict the remaining shelf life of fresh food.



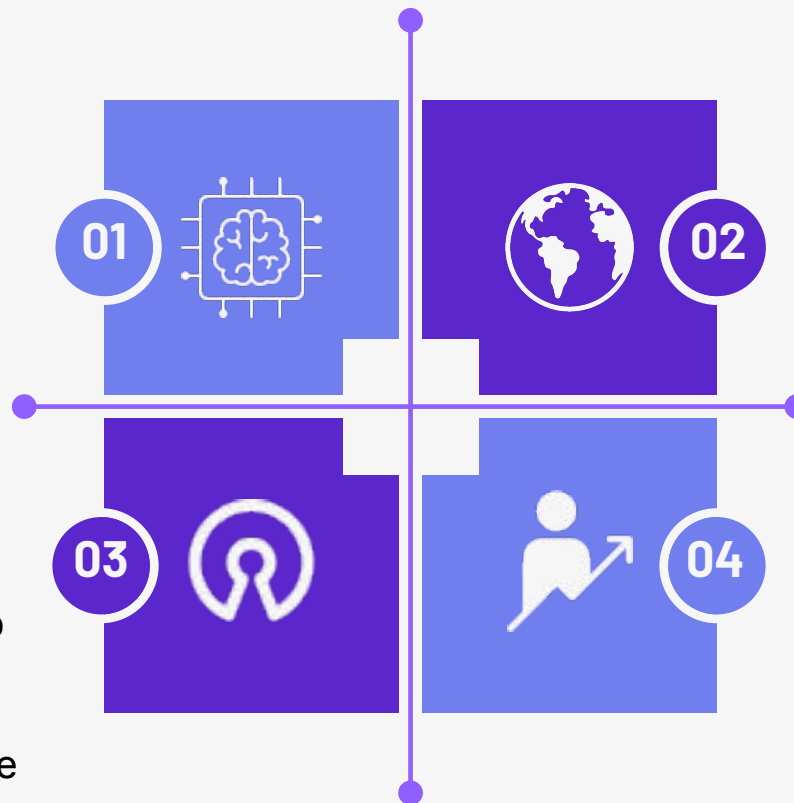
Smart Alerts and Actions

Provide timely alerts and recommend actions such as inspecting or rerouting shipments, adjusting storage conditions, or prioritizing products for sale.

How to Win Reboot 2026

Use AI Responsibly

Innovative, ethical and privacy-respecting uses of AI to solve problems that were previously intractable



Open Source and a Digital Public Good

Solutions that are on track to be digital public goods, that work towards the Sustainable Development Goals and follow the Digital Principles

Relevance to societal challenges

Addresses a pressing need in society, delivering change in a dimension previously unexplored

Scalable, with real-world Impact

Solutions with a tangible roadmap for growth, public adoption and global impact